

Anirudh Srinivasan

SENIOR ML RESEARCH ENGINEER, BHARATGEN

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Professional Summary

Senior ML Research Engineer specializing in vision-language models, document intelligence, and scalable multimodal systems. Built foundation-model training frameworks, distributed data pipelines, and large-scale LLM/VLM generation systems spanning millions of documents and billion-scale inference workloads. My work has been published at CVPR, ECCV, and ICDAR across grounded document understanding, layout analysis, parsing, and multilingual VLM evaluation. I also write *The Backward Pass*, a first-principles blog series on deep learning, foundation models, and ML systems.

Publications

C=Conference Paper, W=Workshop Paper, P=Preprint, S=Under Submission. * denotes Equal Contribution

[C.4] DocLayout-VL: A Foundational Model for Hierarchical, Open-set, and Promptable Document Layout Segmentation 📄

Srinivasan, A.*, Venna, V. K.*, Bandurupalli, S.*, Raghuvver, R., Gunda, S. M., & Sarvadevabhatla, R. K. (Poster at ECCV '26)

[C.2] REPLICA: An Agentic Framework for Visually Faithful Document Reconstruction 📄

Srinivasan, A.*, Raghuvver, R.*, Venna, V. K.*, Sreevatsa, T., Jain, A., Kukkala, S., & Sarvadevabhatla, R. K. (Oral at ICDAR '26)

[C.3] Patram-Bench: Comprehensive Multi-task, Multi-domain and Multilingual Benchmark for Indian Document Understanding 📄

Srinivasan, A.*, Jena, P.*, Topale, A., Venna, V. K., & Sarvadevabhatla, R. K. (Poster at ICDAR '26)

[C.5] DoCoG: Mask-based Multi-Type Grounded Chain-of-Thought for Document QA 📄

Gunda, S. M.*, Jinka, J. S.*, Rachakonda, H. S.*, Jain, A., Venna, V. K., Srinivasan, A. & Sarvadevabhatla, R. K. (Spotlight at ECCV '26 and Poster at eXCV, MUCG, UniWorld@ECCV'26)

[C.1] [W.1] [W.2] M3Grounder: Mask-Based Multi-Span and Multi-Granular Grounding for Document QA 📄

Venna, V. K.*, Gunda, S. M.*, Jinka, J. S., Rachakonda, H. S., Srinivasan, A. & Sarvadevabhatla, R. K. (Poster at CVPR'26, MMFM5, DataMFM@CVPR'26 & Highlight at GRAIL-V@ CVPR'26)

Work Experience

BharatGen

Hyderabad, India

SENIOR ML RESEARCH ENGINEER MENTORED BY

DR. RAVI KIRAN SARVADEVABHATLA AND DR. GANESH RAMAKRISHNAN

Apr 2025 - Present

• Patram-1.5: A Foundation VLM for Holistic Document Understanding across Multiple Tasks

- Established a new **Pareto frontier for Document VLMs**, ranking #1 across perception (OCR, parsing, key-value extraction, chart/table parsing) and reasoning (abstractive VQA, summarization, CoT, ambiguous/adversarial QA) across multiple benchmarks, outperforming models including Qwen3-VL-8B; introduced **OCR 3.0**, a customizable, prompt-driven parsing framework.
- Designed and ablated the full training curriculum, converging on **PT → Mid-Training → SFT → Long-Context SFT → RLFT**, including data curriculum for captioning, OCR, VQA, JSON/Markdown parsing, and HTML document reconstruction.
- Developed scalable VLM training techniques including **task-balanced sampling**, **task-aware token weighting**, **message-tree attention**, **bidirectional visual-token attention**, **dynamic sequence packing**, and staged **36K+ token / 384-frame long-context SFT with context parallelism**; ran multi-task data-mixing ablations to mitigate catastrophic forgetting.
- Ablated vision encoders (**OpenVision**, **SigLIP2**, **CLIP**, **MetaCLIP**) and language decoders (**OLMo3**, **Qwen3**, **Gemma4**) to identify the optimal architecture combination.

• Patram-Layout: A Foundation VLM for Reimagining Document Layout Understanding

[C.4]

- Reframed layout understanding as **promptable, open-set hierarchical segmentation**, generating **layout trees + 80M masks**
- Designed an **end-to-end VLM + segmentation model** with **hierarchy-aware grounding (HGC)** for layout parsing.
- Trained with **LM + Dice/BCE + structural losses (containment, overlap)** on **2.5M documents / 400K+ labels**

• Patram-Grounder: A VLM for Grounded Document Intelligence

(Presented at IndiaAI Impact Summit '26)[C.1][C.5]

- Built an **end-to-end VLM + segmentation model** for DocVQA that generates **answers with interleaved grounding tokens** and **pixel-level masks** (phrase/line/block), replacing coarse bbox grounding with fine-grained segmentation
- Introduced **multi-span, multi-granular & multi-type grounding** via **[GROUND] tokens + learned prompt embeddings**, enabling joint reasoning over **textual + graphical elements** with **hierarchical enclosure & interaction modules**
- Scaled training on **200K–325K documents & 1.5M–2M grounded QA pairs** with **Dice+BCE + bleed suppression + hierarchy losses / DPO**, achieving **SOTA grounded DocVQA** with strong gains in both answer accuracy and grounding fidelity

• Patram-Core: Training & Post-Training Framework for Vision-Language Foundation Models

[C.4] [C.3] [C.5]

- Built a **fully config-driven VLM stack** for PT, SFT, 36K long-context SFT, grounding, and RLFT, supporting **Qwen2.5/3**, **Gemma3**, **Molmo1/2**, **OLMo/OLMo3**, **10 vision encoders**, component-wise freeze/init, encoder/decoder swaps, prompt conditioning, and modular **VLM+SAM** with MLP/Transformer connectors.
- Engineered the multimodal data stack with **47+ datasets**, streaming, multi-image/video inputs, hierarchical **sqrt-size/task-balanced mixtures**, cost-aware sampling/curricula, document augmentation, task/answer-length weighting, and **deterministic 2-D token/crop knapsack packing**.

- Built **custom multi-task training/evaluation callbacks** for task-specific generative evaluation, distributed-safe decoding, metric aggregation, W&B qualitative tables, metadata-wise breakdowns and sampling sweeps; added callbacks for **MFU/token throughput/memory**, profiling/NVTX, validation visualizations, checkpoint metadata/sidecar evaluation, graceful stopping, and exact mixture-state resume.
 - Scaled training via **FSDP2/HSDP, DeepSpeed ZeRO-3, DDP, and DP×Ulysses**, with bf16, configurable/selective activation checkpointing, invariant global-batch sizing, sharded checkpoints, **elastic multi-node recovery**, synchronized OOM skipping, and topology-independent exact streaming-data resume.
 - Optimized training with **sparse supervised-token logits (63% lower peak memory, 1.6× faster on H100)**, FlexAttention block masks, fused AdamW, tuned prefetching, and profiling; added component-aware **LoRA/DoRA/PEFT, GRPO/DPO + configurable rewards + colocated vLLM**, HF/vLLM VLM+SAM serving, OpenAI-compatible APIs, and batched inference with **2.9× throughput**.
- **Patram-Parse: An Agentic Document Parsing Framework** (Presented at Impact AI Summit '26)[C.2]
 - Built an **agentic document parsing workflow** generating **Fid-HTML** with **93% visual fidelity** outperforming Gemini 2.5 Pro.
 - Designed coordinated agents: **Segment (hierarchical layout tree), Router + Specialist Agents (text/table/image/form/list/eqn), Reading-Order Agent**, and **Reflection Agent**, achieving **0.97 routing accuracy / 0.96 DOM validity**
 - Achieved **SOTA 0.86 overall score** on VFDR-Bench (5K docs, 15 langs, 15 domains) with **85% physical structure fidelity**
- FOUNDING ML ENGINEER MENTORED BY
DR. RAVI KIRAN SARVADEVABHATLA AND DR. GANESH RAMAKRISHNAN Jun 2024 - Mar 2025
- **Patram-Bench: A Comprehensive Indian Document Understanding Benchmark** (Presented at BharatGen Summit '25)[C.3]
 - Built a **multilingual document benchmark** with **52,607 documents, 23 languages, 13 domains, 71 document types** and **52 tasks**, comprising **2.4M+ QA annotations** across OCR, parsing, layout, and multi-hop DocVQA
 - Designed **Patram-Syn: a persona-driven synthetic pipeline** with **200K+ personas** and a **5-stage generation stack** (retrieval → structured code → multilingual rendering → human validation) for charts, tables, HTML and diagrams
 - Introduced **multilingual robustness metrics (DUCLI, SRS)** and evaluated **29+ VLMs**, exposing **cross-lingual degradation, script sensitivity**, and failures in **layout-intensive, low-resource settings**
 - **Patram-DB: High-Performance Multimodal Database for VLM Pipelines**
 - Built a PostgreSQL + FastAPI unified data platform for large-scale VLM training, managing **50M+ document images** and **250M+ multimodal annotations** via a sharded architecture with declarative partitioning.
 - Engineered a high-concurrency ingestion pipeline achieving **65k+ rows/sec** peak throughput using async multi-threaded PostgreSQL COPY, UNLOGGED staging tables, and batched transactions.
 - Achieved **<1.5 ms P99 query latency** by implementing PgBouncer connection pooling, query partition pruning, and JSONB with GIN/B-tree composite covering indexes for open-set metadata retrieval.
 - Guaranteed real-time referential consistency and audit-trailed versioning across distributed storage using C-level triggers, materialized views, and TOAST management to prevent bloat from massive OCR payloads.
 - **Patram: India's First VLM built-from-scratch for Document Understanding** (Launched at BharatGen Summit '25)
 - Built and open-sourced **Patram-7B**, a document VLM with a **CLIP-style ViT + gated projector + OLMo-7B [HF Link]**.
 - Curated **BharatDocs: 20M+ documents / 250M+ pages** across **English + 22 Indian languages**.
 - Pre-trained on **1M+ document images** for captioning and instruction-tuned on **500K VQA annotations** spanning **20+ task types**.
 - Built a multi-stage **Synthetic QA pipeline** for extractive/abstractive QA generation, followed by verification and deduplication.
 - Extended Patram for **VQA, multi-turn QA, key-value extraction, CoT, OCR, chart, table and flowchart understanding**.
 - Achieved **#1 Text-Rich Document VQA (91.16)** and **#2 overall (86.39)**, outperforming **Qwen3-VL-8B, InternVL3-8B, Gemma-3-12B, and Molmo-7B** across key document benchmarks; scored 92.1 on Patram-Bench-VQA
 - Deployed for **SBI Life's medical insurance workflow**, achieving **92% KVE F1** on pathology/healthcare reports after LoRA fine-tuning.
 - **e-vikrAI: Indic VLMs for e-commerce** (Presented at IMC '24)
 - Built an **e-commerce VLM** for auto-cataloging, generating **SEO-optimized titles**, hierarchical categories, descriptions, and metadata (**price, size, color**) directly from product images.
 - Built a scalable e-commerce scraper for **22.8M+ products**, covering category/product/metadata/image pipelines with **hierarchical link extraction**, anti-bot/CAPTCHA handling, rotating proxies, retry/backoff, checkpointed resume, and high-throughput ingestion via **async I/O, multiprocessing, and Scrapy/Twisted**.
 - Developed a PoC mobile app demonstrating end-to-end cataloging with **multilingual translation** and **text-to-speech (TTS)** for improved accessibility.
 - Reduced cataloging time from **10 min/product** manually to **6 sec/product**, enabling scalable seller onboarding and simplified usage for **rural/non-English-speaking users**.

Pertinent Skills

Programming ML / VLM Stack Optimization Tools

Python, CUDA, C++, JavaScript, DBs (faiss, mongodb, postgresql)
 PyTorch, OpenCV, HF transformers, HF datasets, webdataset, TRL, deepspeed, accelerate, NCCL, slurm
 triton, bitsandbytes, flash-attn, xformers, vLLM, sglang, ollama, cuML, cuDF, ONNX, TensorRT
 git, linux, bash, wandb, gradio, streamlit, nsys, ncu, docker, fastapi

Education

Indian Institute of Technology, Hyderabad

CGPA: 8.76/10.00

BACHELOR OF TECHNOLOGY IN COMPUTER SCIENCE AND ENGINEERING : **INSTITUTE GOLD MEDAL FOR EXCELLENCE IN ACADEMICS AND CO-CURRICULAR ACTIVITIES (RANK 1 OUT OF 1103)** July 2020 - May 2024

Key Coursework: Data Structures, Algorithms, DBMS, Applied Statistics, Data Science Analytics, Convex Optimization, Markov Chains, Neuroscience & BCI, Foundations of ML, Deep Learning, Image Processing, Video Content Analysis, NLP, Tensors, Quantum Computing

Selected Scholastic Achievements

- 2026 **Spotlight @ ECCV** for DoCoG: Mask-based Multi-Type Grounded Chain-of-Thought for Document QA
- 2026 **Oral @ ICDAR** for REPLICA: An Agentic Framework for Visually Faithful Document Reconstruction
- 2024 **Institute Gold Medal** for Excellence in Academics and Co-curricular Activities (Rank 1 out of 1103), IIT Hyderabad
- 2024 **Gold Medalist**, IEEE Signal Processing Cup, ICASSP '24, South Korea
- 2023 **Silver Medalist**, ISRO Mid-Prep Event, *Inter IIT Tech Meet 11.0*, IIT Kanpur
- 2022 **Silver Medalist**, Bosch Mid-Prep Event, *Inter IIT Tech Meet 10.0*, IIT Kharagpur
- 2021 **Academic Excellence Awardee**, 13th Foundation Day of IIT Hyderabad
- 2020 **100.000000 percentile** in JEE Mains Chemistry
- 2019 **All India Rank 333**, Kishore Vaigyanik Protsahan Yojana (KVPY) Scholar, SA Stream

Volunteering and Community Service

- 2026 **Reviewer @ AAAI '27**, for submissions pertaining to GUI Agents & Large Multimodal Models
- 2026 **Reviewer @ ICDAR '26**, for submissions pertaining to Document Layout Analysis & Document Reconstruction
- 2025 **Reviewer @ AAAI '26**, for submissions pertaining to VLM Benchmarks
- Fall '23 **Head Teaching Assistant @ Dept. of CSE, IIT-H**, Data Structures lectured by [Dr. Maria Francis](#) & [Dr. M.V. Panduranga Rao](#)
- Fall '22 **Teaching Assistant @ Dept. of CSE, IIT-H**, Intro to Programming lectured by [Dr. Karteek S](#) & [Dr. Ramakrishna Upadrasta](#)

Exploratory Undergrad Research

- IEEE Signal Processing Cup 2024 (ICASSP) – ROBOVOX**: Secured 1st place globally; built a far-field speaker recognition system under real-world noise conditions (ambient/internal noise, reverberation, distance, babble, angle variability); developed baseline via score fusion of SEResNet and ECAPA-TDNN, and improved with ERes2Net + augmentations (RIR, MUSAN, speed) and adaptive s-norm scoring; achieved minDCF 0.67, EER 8.93% (vs baseline: 0.7096, 9.59%) under [Dr. Sri Rama Murty Kodukula](#)
- Quantum Methods for Vehicle Routing Problem (VRP)**: Surveyed classical (LP via CPLEX) and quantum (CQM via D-Wave) approaches with focus on Multiple Vehicle Visits (MVV); implemented spherical q-means in Qiskit for Multi-Depot VRP (MD-VRP), achieving DBI 0.742 and improved routing cost across 13 datasets; demonstrated superior CQM runtime and explored QUBO formulations under [Dr. M.V. Panduranga Rao](#)
- LiDAR Semantic Segmentation for Autonomous Driving**: Built an end-to-end pipeline; enhanced RangeSeg with bicubic and Lanczos interpolation, improving performance on SemanticKITTI, SemanticPOSS, nuScenes; proposed SRGAN-based upsampling for challenging scenarios under [Dr. C. Krishna Mohan](#)
- Neuro-Fuzzy Clustering for Gaussian Mixture Models (GMM)**: Implemented ANN-based reformulation of Expectation Maximization and developed Neuro-Fuzzy C-Means in Python using DEAP; achieved +11.8% improvement in Silhouette Score over GMM under [Dr. Kishalay Mitra](#)
- Adversarial Robustness for IR2Vec**: Performed adversarial training to mitigate targeted attacks; generated adversarial examples using FGSM and explored Discrete Adversarial Manipulation of Programs (DAMP) under [Dr. Ramakrishna Upadrasta](#)
- MotionGPT for Yoga Postures**: Extended MotionGPT; built MoCap pipelines (C3D, AMASS, SMPL, ASF, AMC); trained on in-house data under [Dr. Mohan Raghavan](#)

Positions Of Responsibility and Extracurriculars

- Mentor, Epoch (AI-ML & Data Science Club) @ IIT Hyderabad** ('23-'24)
- Overall Head, Epoch @ IIT Hyderabad** ('22-'23)
- Deputy Contingent Leader @ Inter IIT Tech Meet 11.0** ('22-'23)
- ML Lead, Google Developers Student Club @ IIT Hyderabad** ('22-'23)
- Student Head, Sunshine UG Mentorship Program @ IIT Hyderabad** ('22-'23)
- Core Member, Epoch @ IIT Hyderabad** ('21-'22)
- Core Member, Robotix @ IIT Hyderabad** ('21-'22)
- Campus Tour Coordinator @ Alumni Cell, IIT Hyderabad** ('23-'24)
- DUGC Student Member @ CSE, IIT Hyderabad** ('21-'24)
- Class Representative @ CSE, IIT Hyderabad** ('21-'22)
- Social Media Manager @ CSE, IIT Hyderabad** ('21-'24)
- UG Mentor @ Sunshine, IIT Hyderabad** ('21-'22)
- Workshop Organizer, Ethical Hacking @ Elan & Nvision, IIT Hyderabad** ('20-'21)
- Silver Medal, ISRO Mid-Prep @ Inter IIT Tech Meet** ('23)
- Silver Medal, Bosch Mid-Prep @ Inter IIT Tech Meet** ('22)
- Silver Medal, Freestyle Duo (Whistle + Beatbox) @ Milan** ('23)
- Gold Medal, Short Film Competition @ Milan** ('22)
- Gold Medal, Institute Cricket League @ IIT Hyderabad** ('21)
- Bronze Medal, Cricket @ Diesta** ('21)
- Silver Medal, Table Tennis & Tennis @ Diesta** ('21)
- Silver Medal, Cricket & Basketball @ Milan** ('22-'23)
- Bronze Medal, Swimming @ Milan** ('22)